



2201 321 Veterinary Virology

Order *Reovirales* and Family *Bunyaviridae*

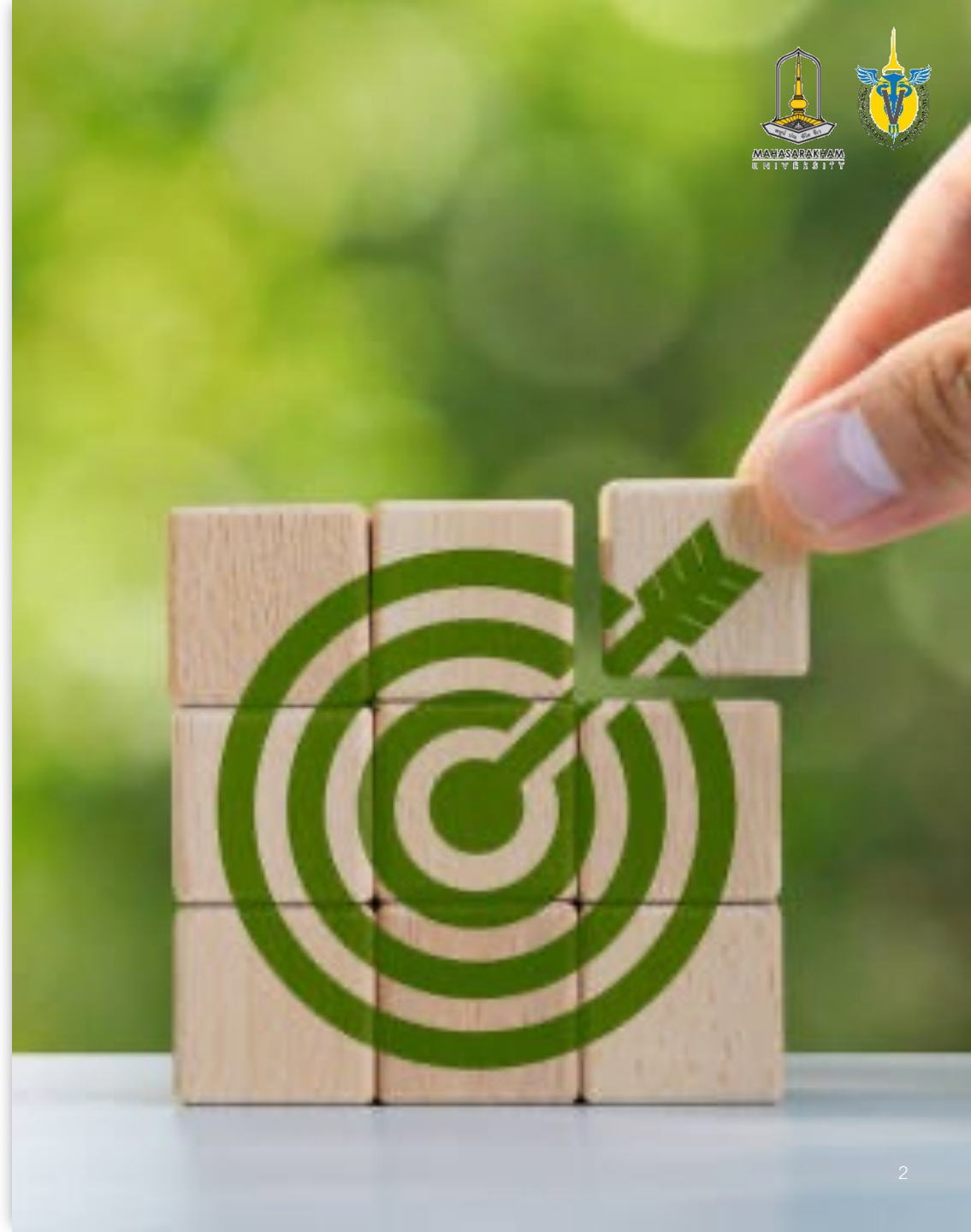
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Mahasarakham University

วัตถุประสงค์

- นิสิตมีความรู้ความเข้าใจเกี่ยวกับไวรัสและโรคที่เกิดจากไวรัสใน Order *Reovirals* และ Family *Bunyaviridae* ดังนี้
 - ลักษณะและชีววิทยาของไวรัส
 - การก่อโรคในสัตว์ พยาธิกำเนิดและอาการแสดงทางคลินิก
 - ตัวอย่างเพื่อชั้นสูตรทางห้องปฏิบัติการ วิธีการ และเทคนิคที่ใช้ชั้นสูตรทางห้องปฏิบัติการ
 - การควบคุมและการป้องกันโรค

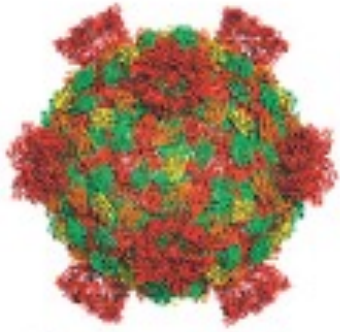


Order *Reovirales*

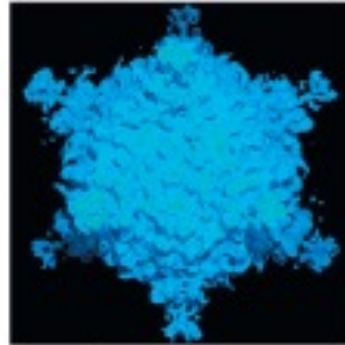
- The family is divided into the two families including *Spinareoviridae* and *Sedoreoviridae*
- These 16 genera include 103 species
 - Family *Spinareoviridae*
 - *Aquareovirus*, *Coltivirus*, *Cypovirus*, *Dinovernavirus*, *Fijivirus*, *Idnoreovirus*, *Mycoreovirus*, *Orthoreovirus* and *Oryzavirus*
 - Family *Sedoreoviridae*
 - *Cardoreovirus*, *Crabreovirus*, *Mimoreovirus*, *Orbivirus*, *Phytoreovirus*, *Rotavirus* and *Seadornavirus*

Reovirus (Respiratory, Enteric, Orphan)

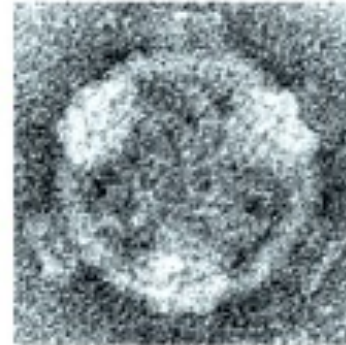
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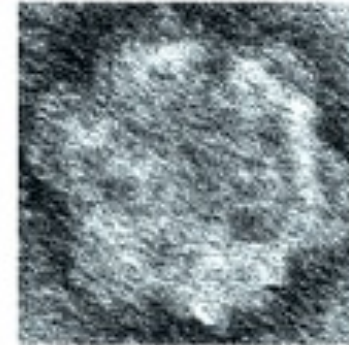
Orthoreovirus



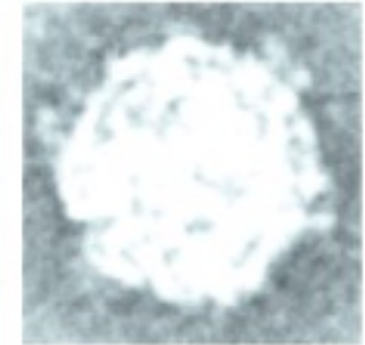
Cypovirus



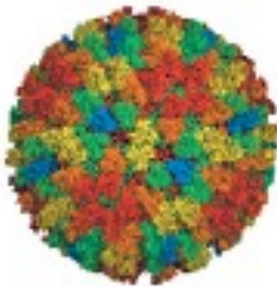
Fijivirus



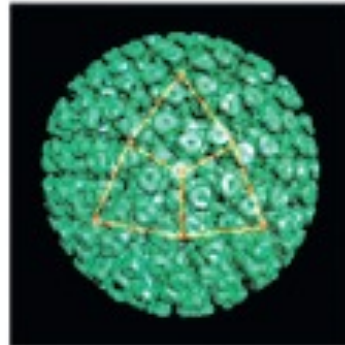
Oryzavirus



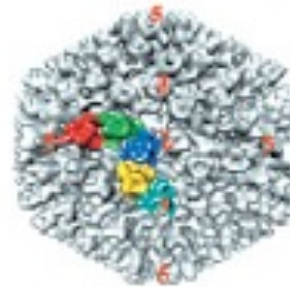
Mycoreovirus



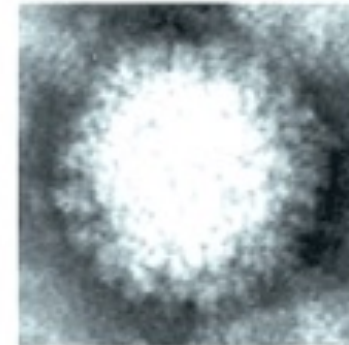
Orbivirus



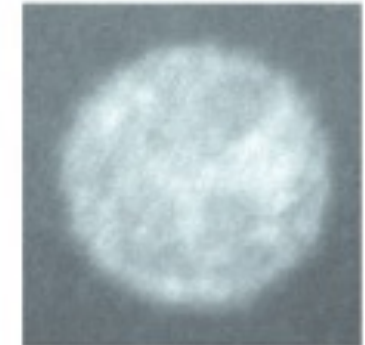
Rotavirus



Phytoreovirus



Coltivirus



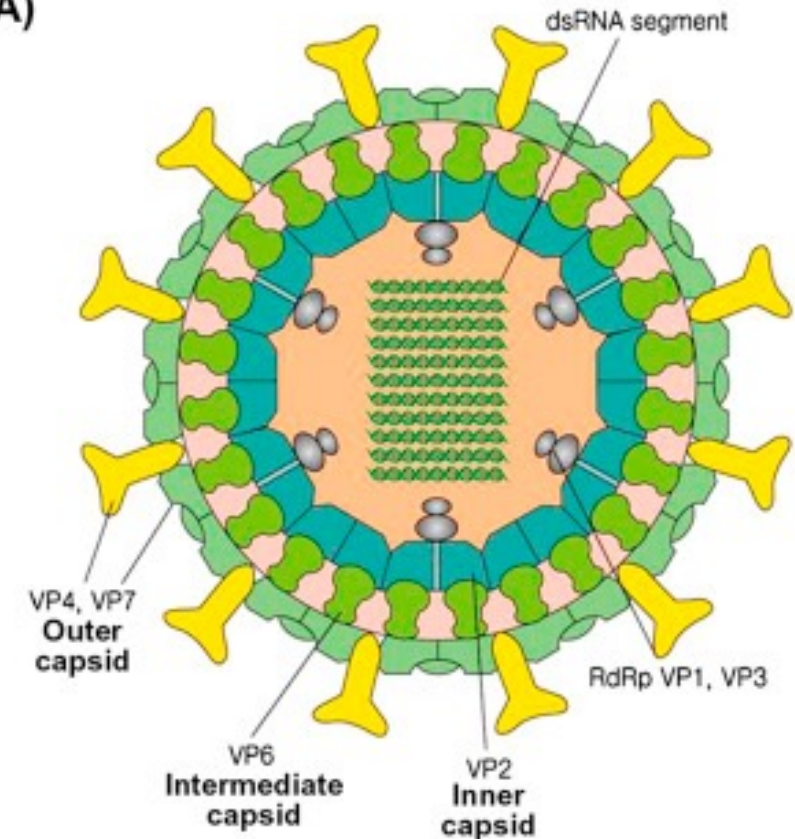
Seadornavirus

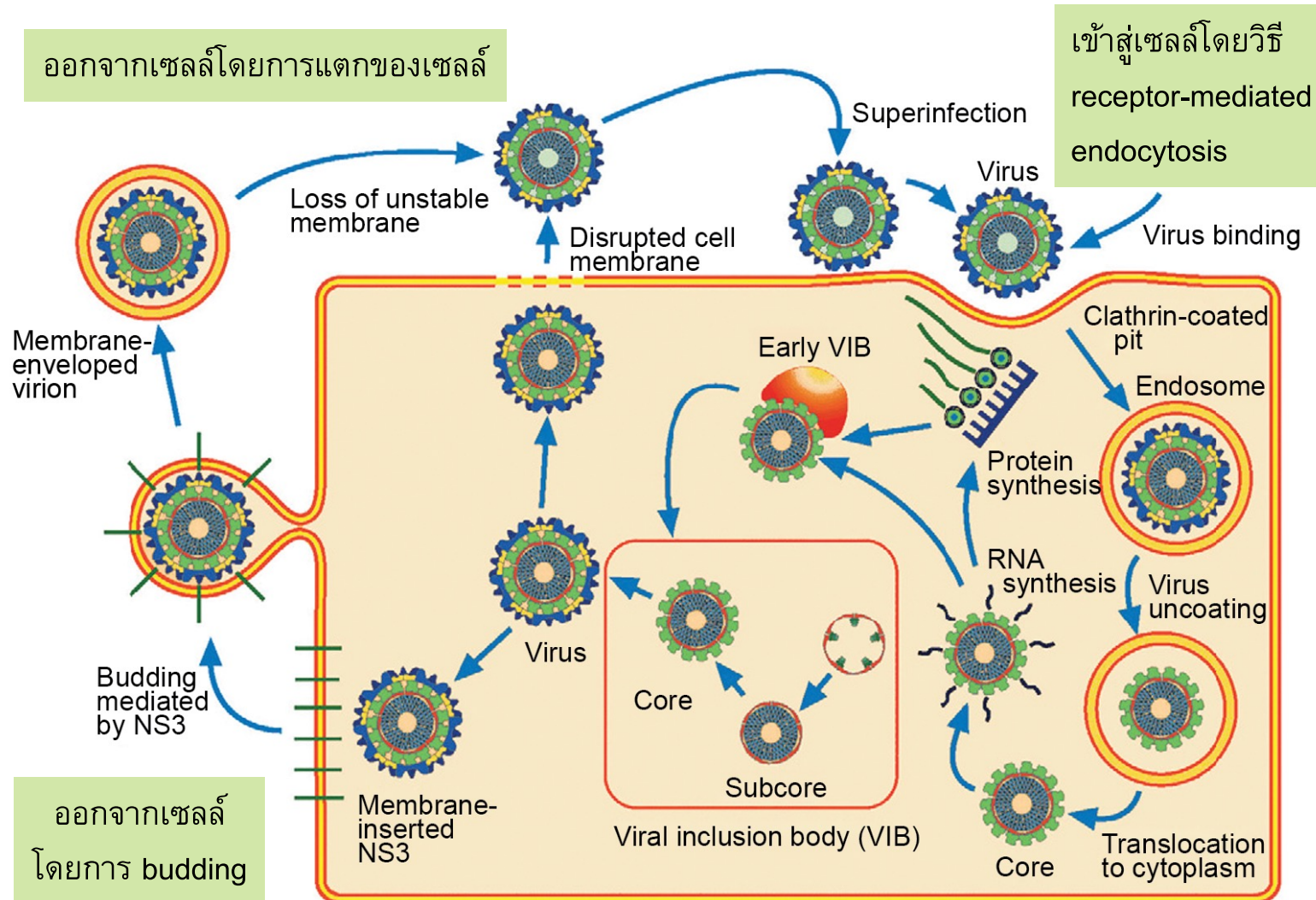
Members of different genera of the family *Spinareoviridae* and *Sedoreoviridae*

Virion properties

- Non-enveloped, icosahedral, 60–95 nm virions composed of 1–3 concentric capsid proteins layers
- Genome: approximately 18–29 kbp of segmented (9–12) linear dsRNA, with. segments ranging from 0.7–5.8 kbp
 - 10 segments (genus *Orthoreovirus* and *Orbivirus*)
 - 11 segments (genus *Rotavirus* and *Aquareovirus*)
 - 12 segments (genus *Coltivirus*, *Seadornavirus*, and some currently designated members of genus *Aquareovirus*)
- Genetic reassortment occurs between viruses within each genus or serogroup
- Host Range: mammals, aquatic animals, birds, reptiles, arthropods, algae, fungi, and plants
- Virion infectivity is moderately resistant to heat, organic solvents (e.g., ether), and non-ionic detergents (depending on both virus strain and detergent)

(A)





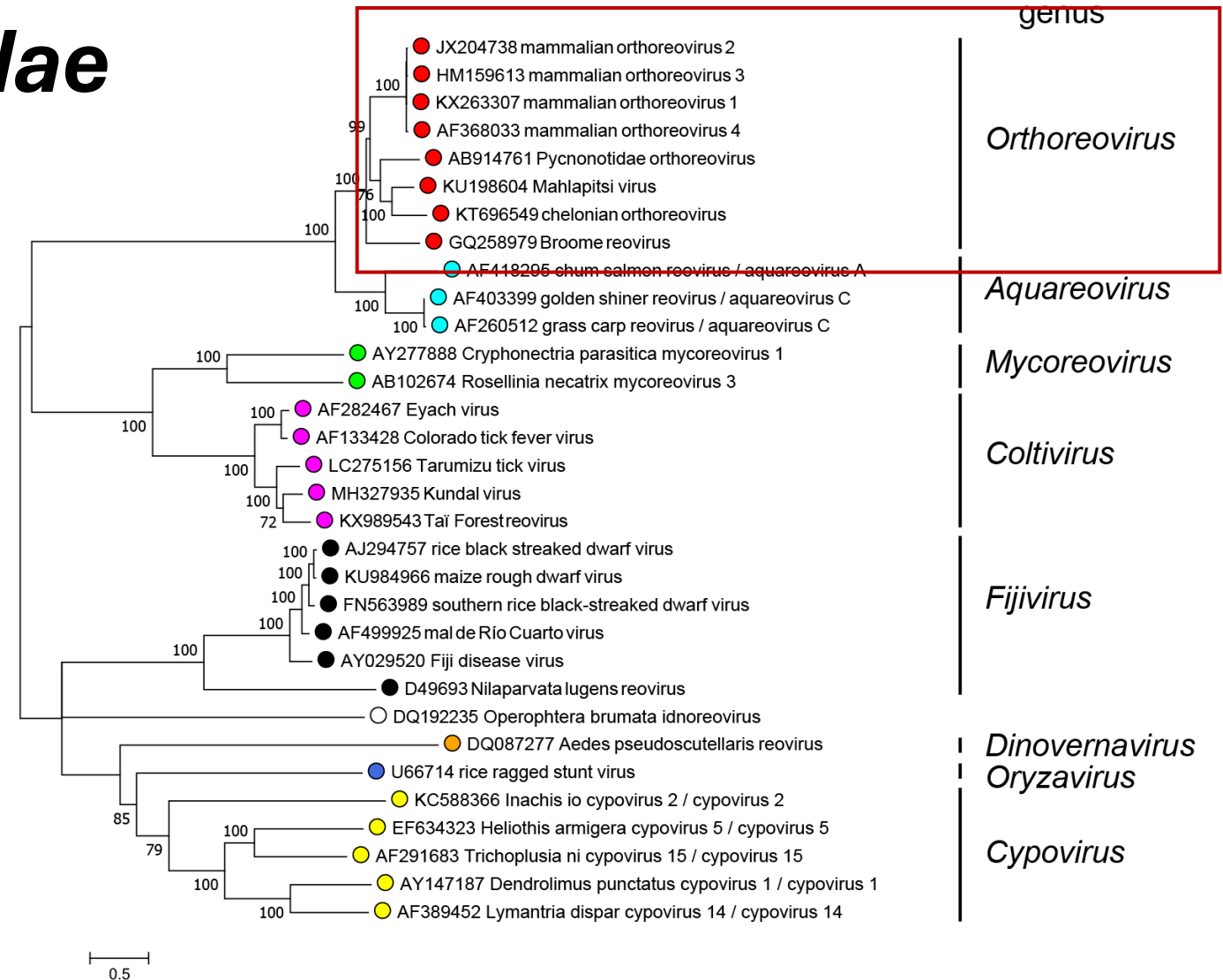
Virus replication

- Replication occurs in the cytoplasm in electron-dense structures called viroplasm or virus inclusion bodies
- Translation occurs from full-length transcribed mRNAs, which possess a 5'-terminal cap but no poly(A)-tail

Family *Spinareoviridae*

- The genera in family *Spinareoviridae* -

- Genus *Aquareovirus*
- Genus *Coltivirus*
- Genus *Cypovirus*
- Genus *Dinovernavirus*
- Genus *Fijivirus*
- Genus *Idnoreovirus*
- Genus *Mycoreovirus*
- Genus *Orthoreovirus***
- Genus *Oryzavirus*





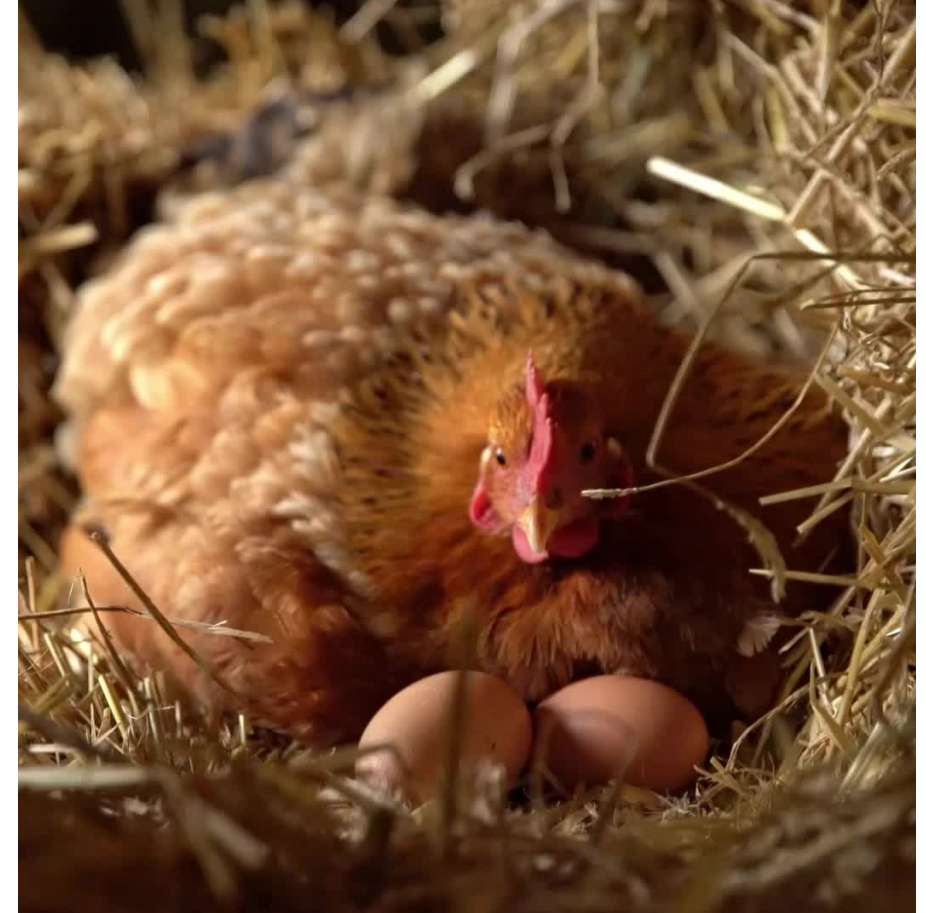
Mammalian orthoreovirus

- Cause an experimental disease syndrome in neonatal laboratory mice - **heptoencephalomyelitis**
 - Jaundice, diarrhea, runting, oily hair and neurologic signs (e.g., ataxia)
- Natural infections in laboratory mice are invariably subclinical
- Surveillance and prevention: *cost!!*
- Infection of primates
 - Associated with both hepatitis and meningitis

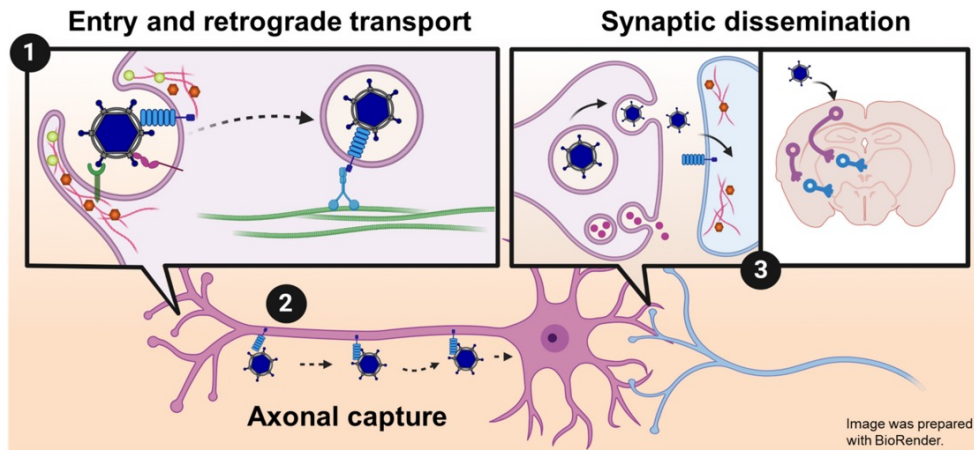


Avian orthoreovirus

- Infection in some chicken, turkey and goose flocks may be subclinical
 - Tenosynovitis and arthritis in chickens
- Other syndromes include;
 - Respiratory disease, gastroenteritis, malabsorption syndrome, hepatic necrosis, hepatitis, myocarditis, hydropericardium, weight loss, runting/stunting and increased mortality
- **Tenosynovitis and arthritis** usually occur in meat producing birds more than 5 weeks of age
 - Morbidity is often 100%, and mortality is usually less than 2%



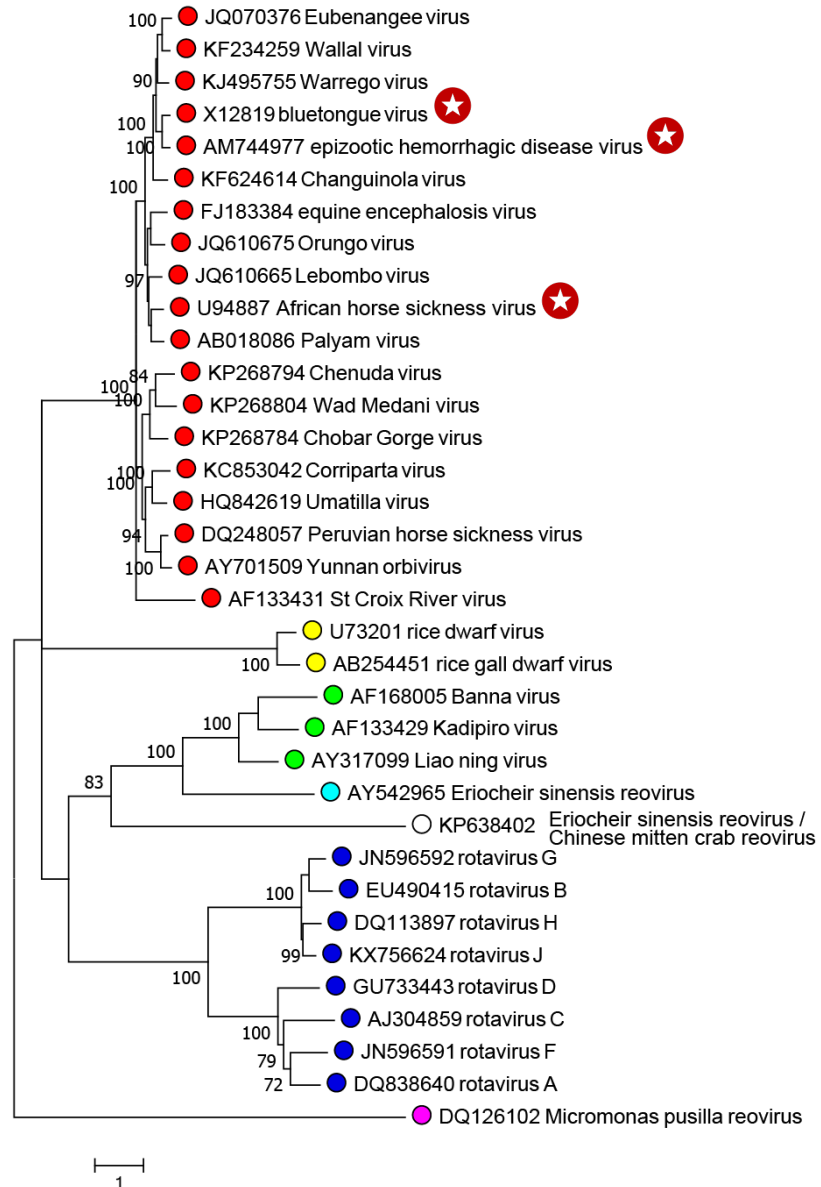
Pathogenesis



- Transmission of orthoreovirus is by an enteric or respiratory route (vertically and horizontally)
 - No arthropod vectors are involved, and infection is restricted to a wide variety of vertebrate species (baboons, bats, birds, cattle, humans, monkeys, sheep, snakes, swine, and rodents)
- After infection, virus spread from an initial portal of entry to sites of secondary replication – lymphoid tissue, GI tract, and respiratory tract
 - Apoptosis of reovirus-infected epithelial cells – inflammation occurs
- Virus can disseminate via the bloodstream; infection of endothelial cells and development of viremia – the virus can spread to multiple locations, such as the heart, liver, intestines, and tendons
- Neuronal transport in mice: the capacity for sustained viremia and efficient neuronal transport differs between MRV serotypes, influencing the development of systemic disease

Diagnosis and control

- Laboratory diagnosis
 - Demonstration of reovirus in affected tissue by virus isolation and RT-PCR
 - Commercially available ELISAs can be used to diagnose reovirus infection; however, results are not definitive
 - Virus neutralization assays can be used to detect type-specific antibodies; however, definitive diagnosis should rely on the detection of virus in affected tissues
- Control and prevention
 - Animal monitoring and wastewater surveillance
 - Immunological strategies; development of effective vaccines is a primary control goal
 - Biosecurity Measures



genus

Orbivirus ★

Phytoreovirus

Seadornavirus

Cardoreovirus

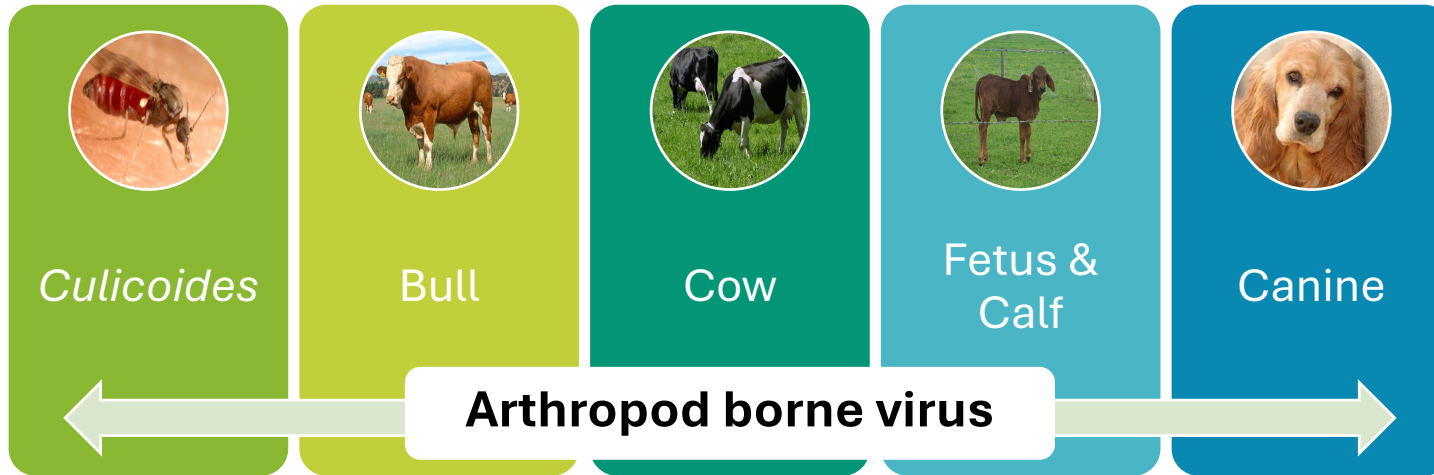
Rotavirus ★

Mimoreovirus

Family Sedoreoviridae

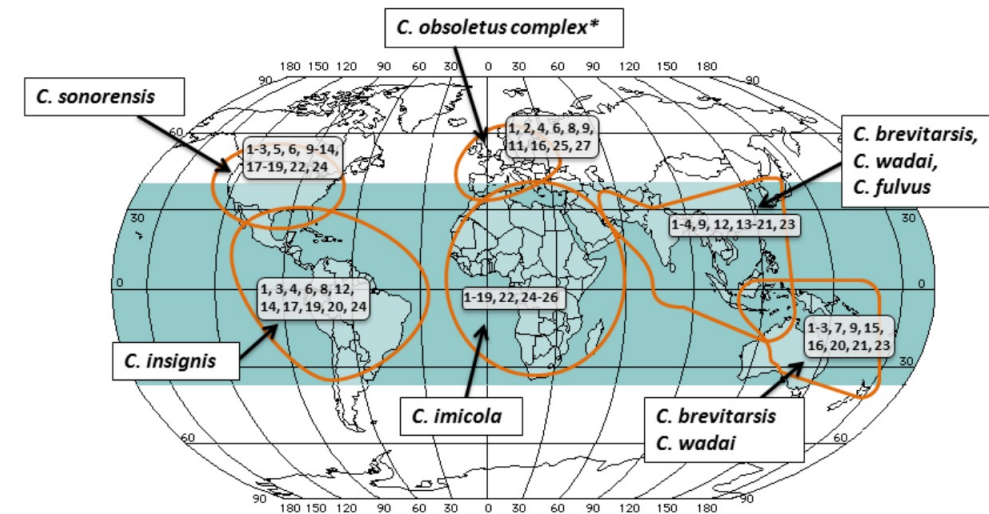
- The genera in family Sedoreoviridae –
 - Genus *Cardoreovirus*
 - Genus *Crabreovirus*
 - Genus *Mimoreovirus*
 - Genus *Orbivirus***
 - Genus *Phytoreovirus*
 - Genus *Rotavirus***
 - Genus *Seadornavirus*

Bluetongue virus



Generalized infection and reproductive failure

Figure 2. Approximate Geographic Distribution of Bluetongue Virus Serotypes and Major Vectors⁵³



* The *C. obsoletus* complex comprises of several other closely related species of midges. This particular complex and its various species is not all inclusive to the European region, but is the primary species in addition to *C. imicola* which resides in the Mediterranean area.

From: Bluetongue Standard Operating Procedures, 2016

Clinical signs: in sheep

- Eye and nasal discharges
- Drooling as a result of ulcerations in the mouth
- High body temperature – 106°F (105-107.5°F)
- Swelling of the head, neck and mouth – muzzle
- Cyanosis of the tongue and mucous membranes (bluetongue)
- Superficial hemorrhages
- Inflammation at the junction of the skin and the horn of the foot – the coronary band
- Lameness
- Respiratory problems – difficulty with breathing and nasal discharge
- Loss of condition and muscular weakness
- Abortion and deformed lambs



Clinical signs: in cattle

- Nasal discharge (mucopurulent)
- Swelling of the head and neck
- Conjunctivitis (runny eyes)
- Swelling and ulceration of the mouth and nose
- Swollen teats
- Excessive salivation
- Depression and tiredness

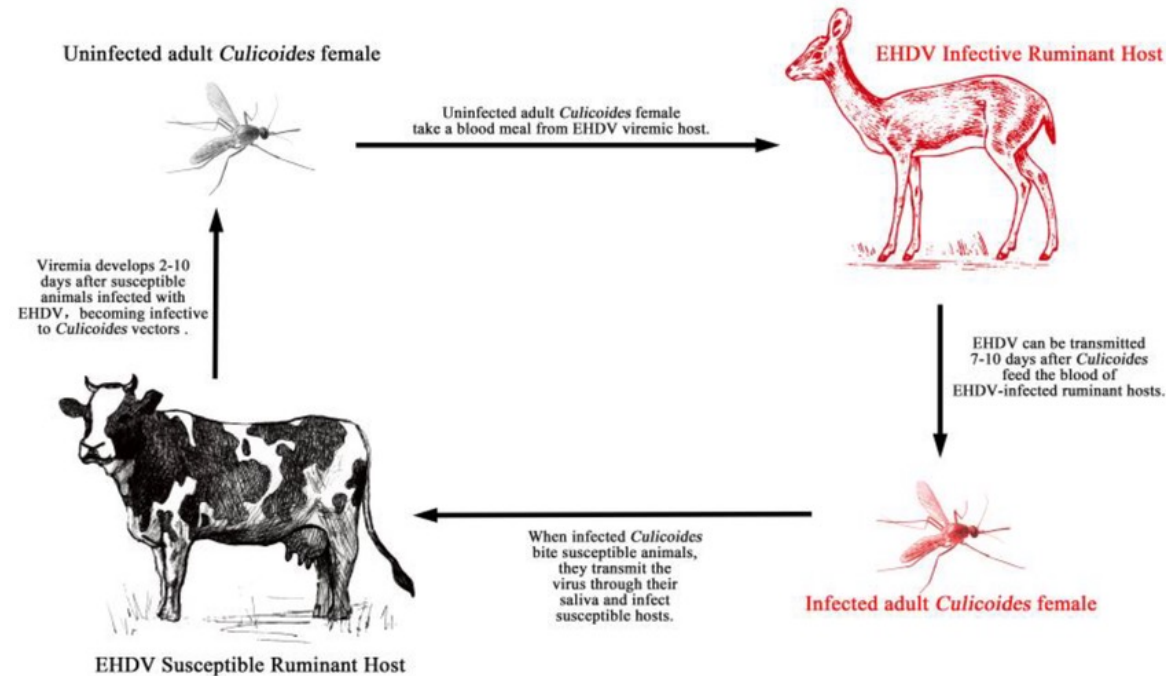


Diagnosis and control

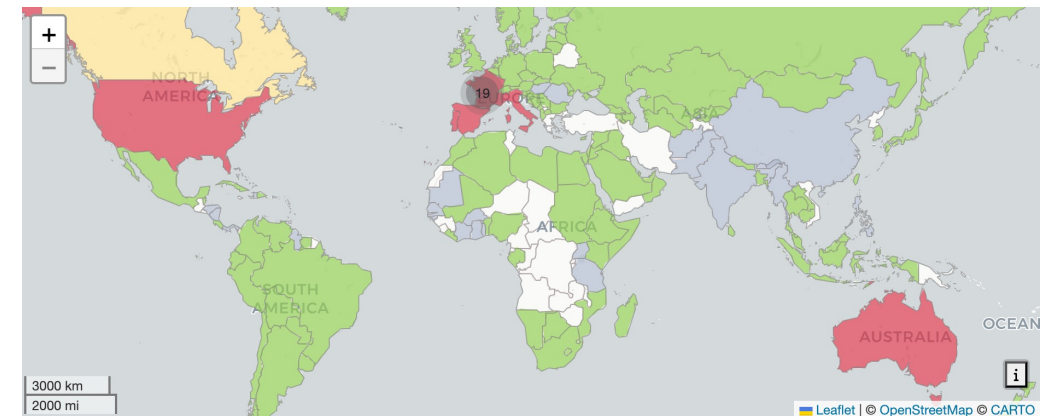
- Diagnosis: antigen or/and antibody detection
 - Virus isolation
 - Polymerase chain reaction (RT-PCR)
 - Complement Fixation (CF)
 - Agar Gel Immunodiffusion (AGID) - false positive
 - Competitive ELISA (C-ELISA)
- Control and prevention: vaccination, vector control



Epizootic hemorrhagic disease virus



- Transmission: the virus is spread by the bites of small, biting midges (*Culicoides* spp.)
- Affected animals: the disease primarily affects wild white-tailed deer but can also infect cattle, sheep, and other wild ruminants
- Geographic distribution: North America, Asia, Africa, and Australia, with outbreaks most frequent in the southern United States



Clinical signs

- Deer: high fever, weakness, and lethargy, death can occur rapidly, sometimes within 48 hours of symptoms appearing
 - Peracute form: severe edema of the head, neck, tongue, conjunctiva, and lungs
 - Acute form: widespread hemorrhages and edema in the mucous membranes, skin and viscera, especially heart and gastrointestinal tract
 - Erosions may be found in the mouth, rumen and omasum, and necrosis in the hard palate, tongue, dental pads, esophagus, larynx, rumen and abomasum
 - Chronic form: growth rings on the hooves or sloughing of the hoof wall, and erosions, ulcers or scars in the rumen
- Cattle: similar to bluetongue, including fever, reduced appetite and milk production, swelling of the conjunctiva and nose, and lameness



Diagnosis

- Laboratory testing is the only definitive way to diagnose EHD
 - Virus isolation and RT-PCR
 - Serological tests
 - Competitive ELISA (C-ELISA) (serogroup specific)
 - Virus neutralization (VN) (serotype specific); the gold standard for the identification and quantification of antibodies against EHDV serotypes present in test samples is the VN test
 - Agar gel immunodiffusion
 - The AGID can detect antibodies from 5–15 days after infection to 2 years or more
 - Complement fixation test (CFT) (serogroup specific)
- EHD is a notifiable disease to the World Organization for Animal Health (WOAH) and should be reported if signs are observed

African horse sickness virus

- **95% mortality after infection of susceptible horses**
 - Mules and donkeys are susceptible to infection
- **Transmission**
 - *Culicoides imicola* and *Culicoides bolitinos* are vectors of the virus in South Africa
 - Highly seasonal: late summer
- **Pathogenesis and Pathology**
 - After the bite of an infected insect, the virus replicates in the local lymph node → spread to other tissues and organs → vascular injury in fatal horse → pro-inflammatory and vasoactive mediators released from virus-infected cells (dendritic cells, macrophages, and endothelial cells)

Clinical Features

- **Pulmonary form** (Dunkop or central form)
 - Severe and progressive respiratory distress and death
 - IP 3–5 days, horses develop fever, increases respiration rate
 - Cough and discharge of frothy fluid from the nostrils
- **Cardiac form** (Dikkop or peripheral form)
 - Fever 3–6 days after incubation period
 - Edema (supraorbital fossae, eyelids, lips, tongue, intermandibular space and laryngeal region) sometimes congestion and hemorrhage in the conjunctiva
 - Mortality rates for such cases may be as high as 50%; death occurs within 4–8 days of onset of fever
- **Mixed form** is seen in some animals: mortality is approximately 70%



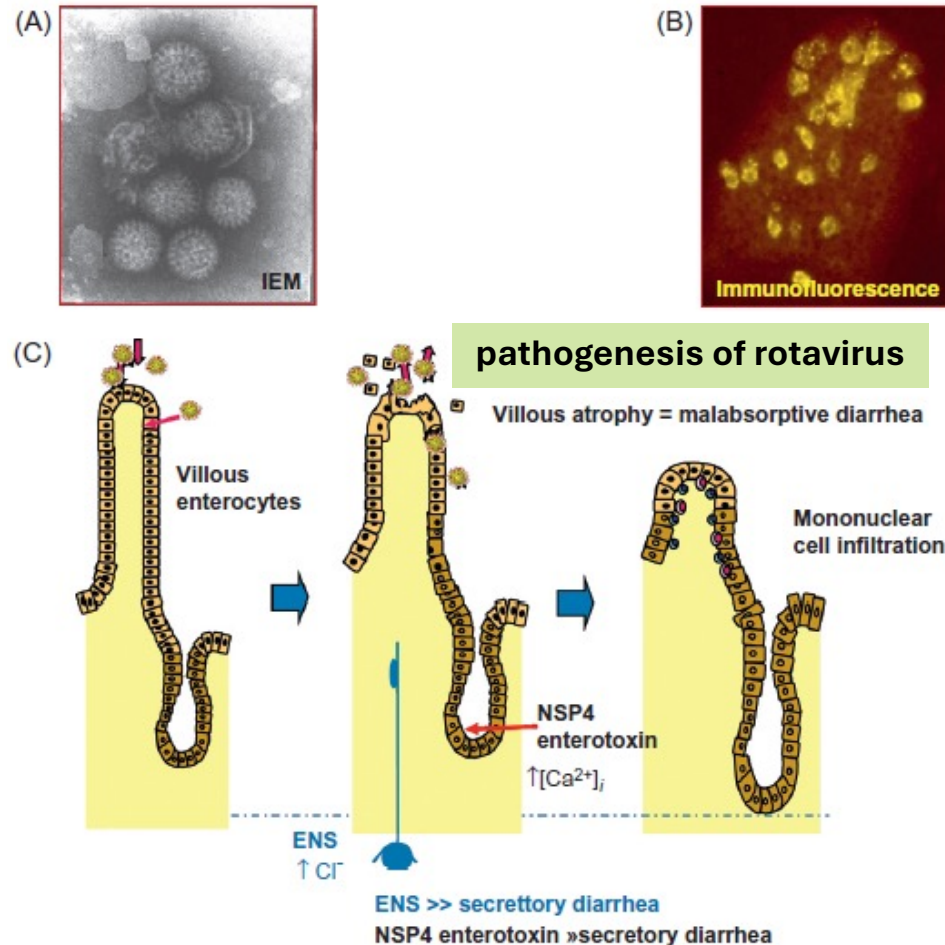
Diagnosis and control

- Laboratory diagnosis
 - Isolated in cell culture
 - Intracerebral inoculation of 2- to 6-day-old mice with blood or a spleen suspension from the suspect animal
 - Neutralization assays
 - RT-PCR assay
- Control and prevention: vaccination with live-attenuated vaccines, vector control through insect-proof housing and repellent use, confinement of equids during peak *Culicoides* midge activity, and strict restriction of animal movement to prevent disease spread
 - For AHSV-free areas, surveillance, quarantine, and reporting of suspected cases to authorities are crucial

Genus *Rotavirus*: infections of mammals and birds

- Disease is usually seen only in young animals
 - 1–8 weeks of age
- Diarrhea in calves, piglets, foals, and lambs
 - “white scours” or “milk scours”
 - Risk factor: ingestion of a large volume of milk, reduced colostrum intake, infections with other enteric pathogens such as *Escherichia coli*, poor hygiene, chilling, and overcrowding
 - Young animals may die as a result of dehydration or secondary bacterial infection
 - Most recover within 3–4 days

Pathogenesis and Pathology



- Rotavirus infections cause intestinal malabsorption and maldigestion by destruction of the terminally differentiated enterocytes lining the tips of the intestinal villi
- Damaged villi become shortened and covered with immature, less differentiated epithelial cells that migrate from the crypts
- Undigested lactose in the milk promotes bacterial growth and exerts a further osmotic effect; both mechanisms contribute to the diarrhea

Diagnosis and control

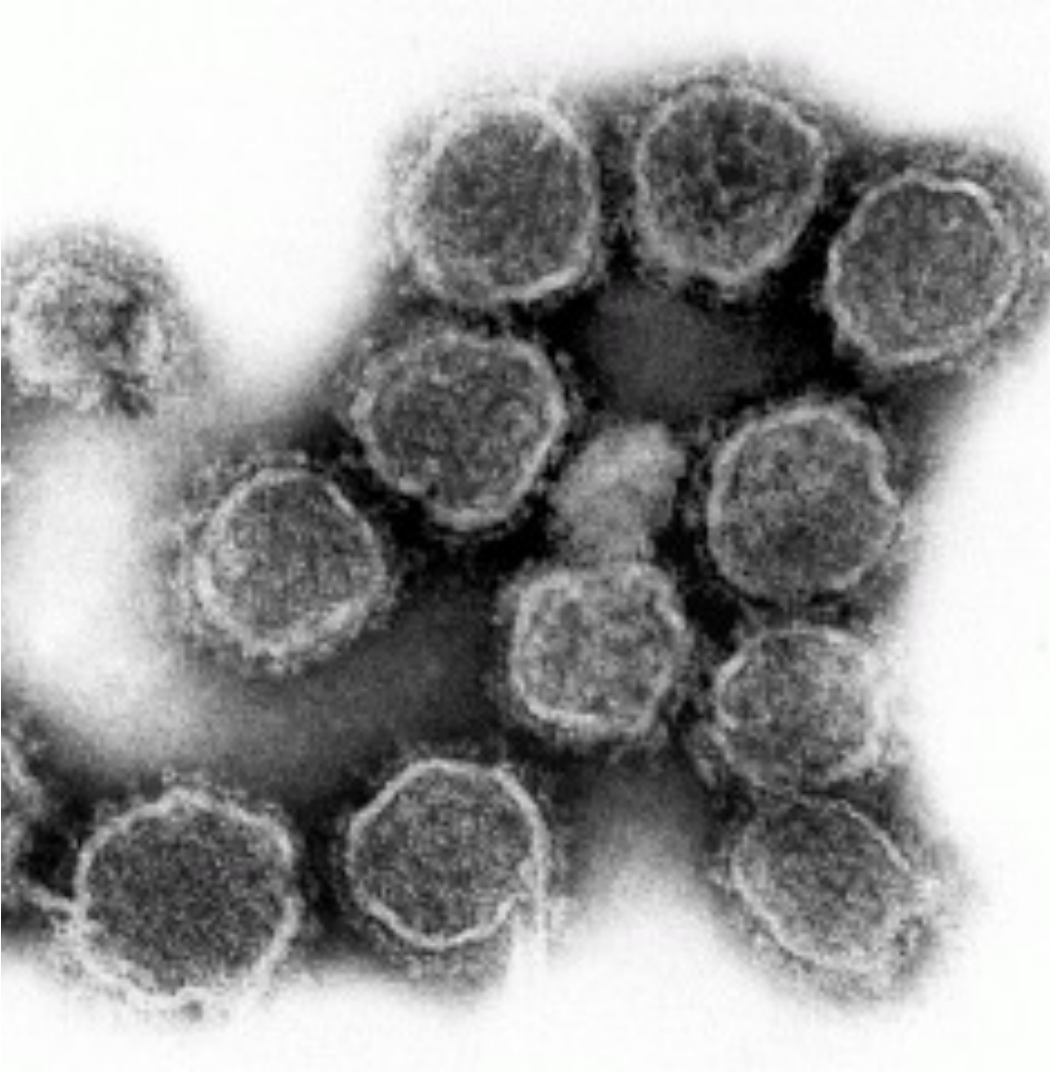
Diagnosis

- Isolate in cell culture
- Electron microscopy
- Enzyme immunoassay
- RT-PCR assay
- Immunofluorescence or immunohistochemistry
- Neutralization tests

Prevention and Control

- Good management for farm animals
 - In domestic mammals, rotavirus antibodies present in immune colostrum and milk are particularly important in protecting neonatal animals





Family *Bunyaviridae*

- **Arthropod-borne (Arboviruses)**
 - Genus *Orthobunyavirus*
 - Genus *Phlebovirus*
 - Genus *Nairovirus*
- **Non-arthropod-borne (Rodent-borne virus)**
 - Genus *Hantavirus*



Order Reovirals and Family Bunyaviridae



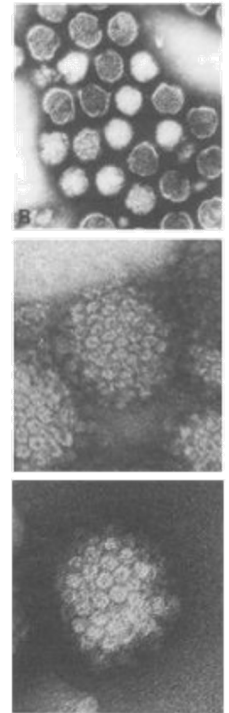
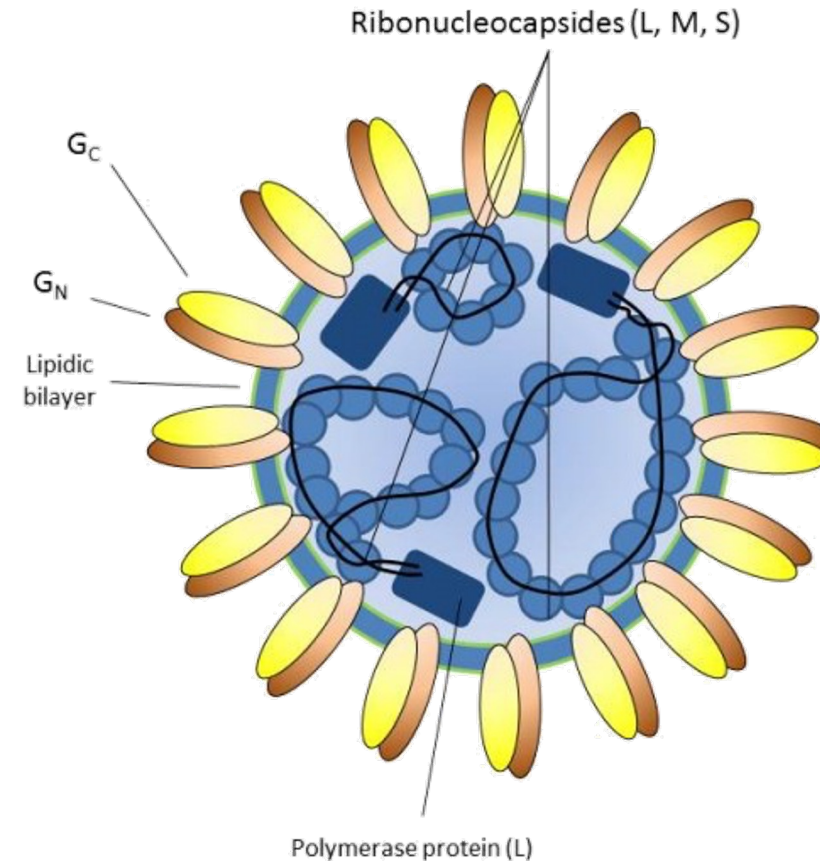
The vectors for virus transmission

- Most are arthropod-borne virus
- Vectors: Mosquito, Tick, Sandfly, Midges
 - **Transovarial transmission:** ไวรัสจะติดเชื้อที่ไข่ของแมลง เมื่อกลายเป็นตัวอ่อนและตัวเต็มวัยจึงติดเชื้อและสามารถติดต่อไปสู่โฮสต์อื่นได้



Virion properties

- Spherical, enveloped virus
- เส้นผ่าศูนย์กลาง 80-120 nm
- Envelope มีไขมัน 2 ชั้น และ glycoprotein peplomer ล้อมรอบ nucleocapsid
- Genome: three unique molecules of negative or ambisense-ssRNA, which total 11–19 kb:
 - L (large)
 - M (medium)
 - S (small)



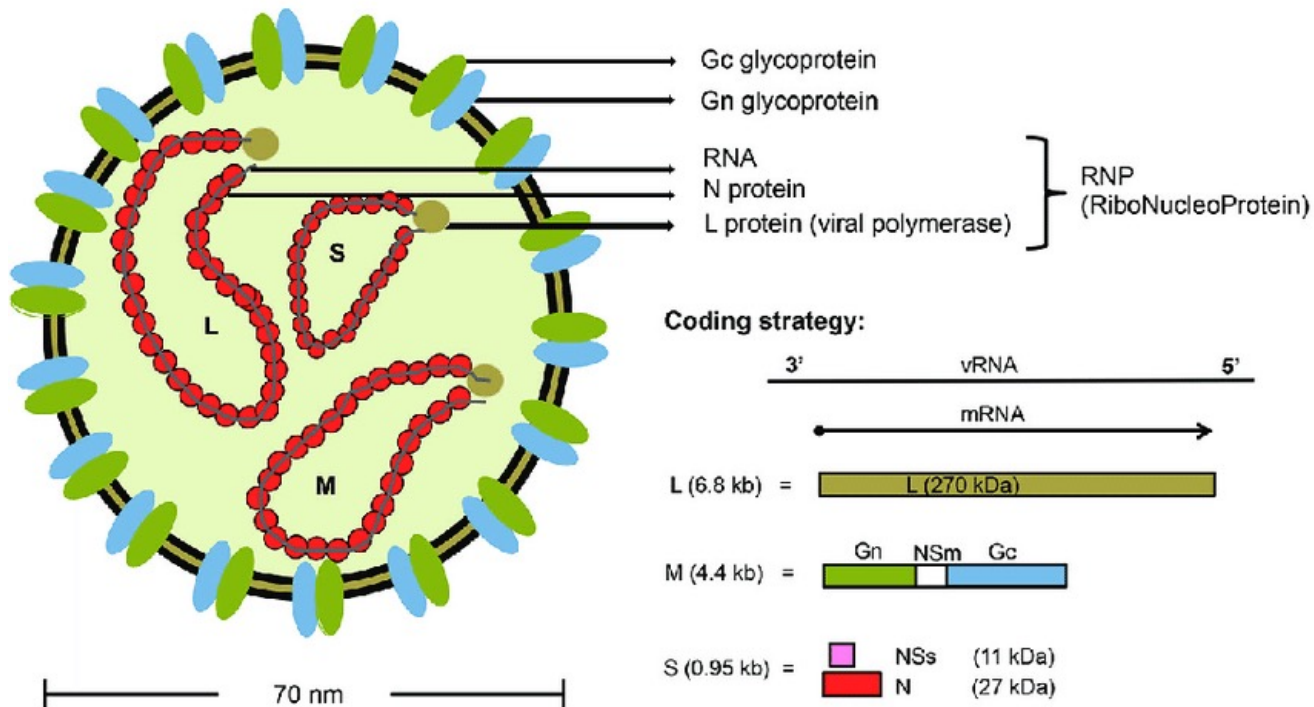
Von Bonsdorf et al., 1975, Jvi

Table 42.1 Pattern of Sizes of Viral RNA Segments in *Bunyaviridae* Genera

Genus RNA segment	<i>Orthobunyavirus</i>	<i>Hantavirus</i>	<i>Nairovirus</i>	<i>Phlebovirus</i>	<i>Tospovirus</i>
L	6.9	6.5	12.2	6.4	8.9
M	4.5	3.6	4.9	3.5	4.8
S	1.0	1.7	1.7	1.7	2.9
Total	12.4	11.8	18.8	11.6	16.6

Size: kb

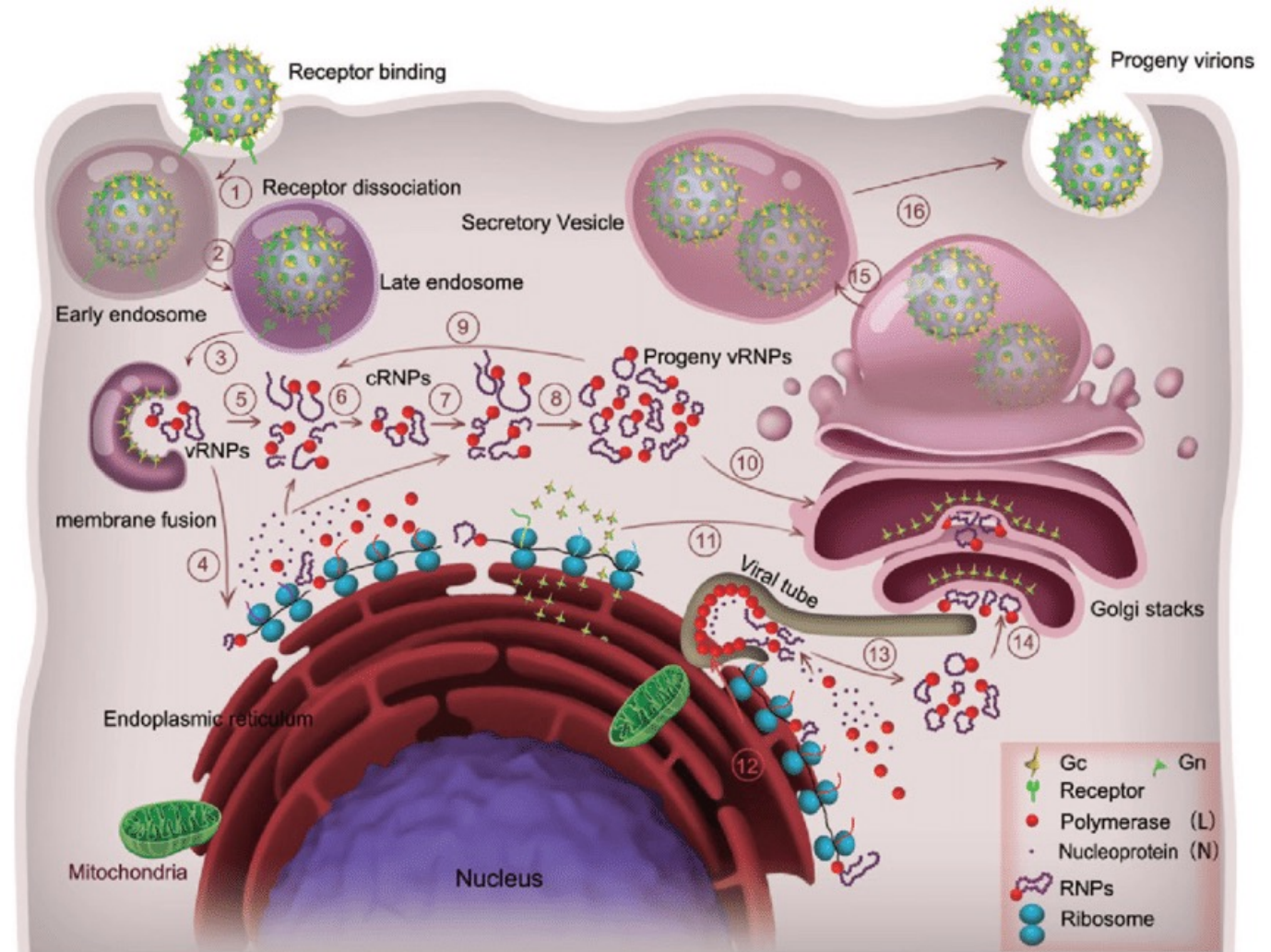
Data From: <https://basicmedicalkey.com/>



- Viral protein: ประกอบด้วยโปรตีนโครงสร้าง 4 ชนิด คือ
- Transcriptase (L), Nucleoprotein (N) และ glycoprotein 2 ชนิด ได้แก่ G1(Gc) และ G2 (Gn)
- G1(Gc) และ G2 (Gn) เป็นส่วนของ epitope บนผิวอนุภาคไวรัส

Virus replication

- ไวรัสเพิ่มจำนวนใน cytoplasm ของ cell host
- maturation โดย budding เข้าไปใน Golgi vesicle
- ถูกปลดปล่อยโดยการเชื่อมของ vesicle membrane กับ เยื่อหุ้มเซลล์
- หลุดออกนอก host โดยกระบวนการ exocytosis



From: Bunyavirales ribonucleoproteins: the viral replication and transcription machinery (Sun et al, 2018)

TABLE 22.1 Family *Bunyaviridae*: Major Pathogens of Animals and Humans

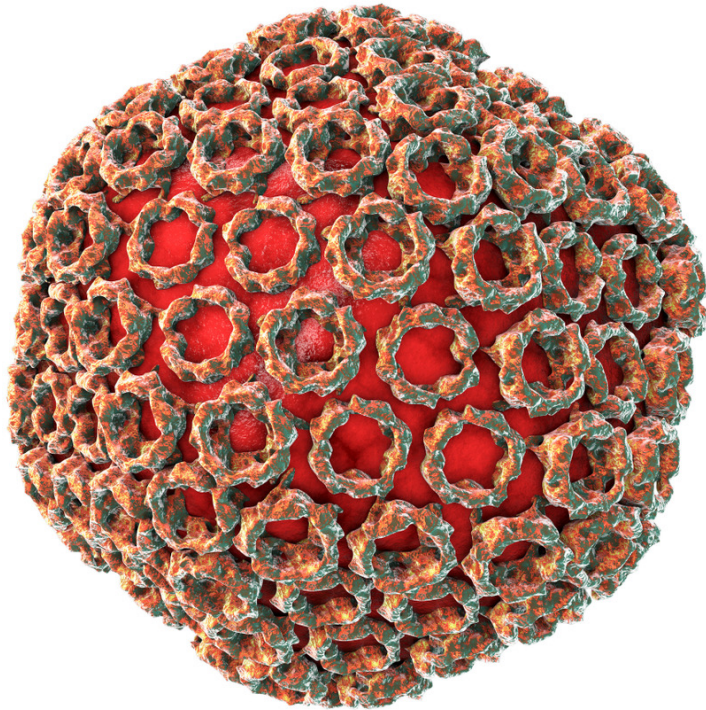
Genus	Virus	Geographic Distribution	Arthropod Vector	Target Host Species or Amplifier Host	Disease in Animals	Disease in Humans
<i>Phlebovirus</i>	Rift Valley fever virus	Africa	Mosquitoes	Sheep, cattle, buffalo, humans	Systemic disease, hepatitis, abortion	Flu-like illness, hepatitis, hemorrhagic fever, retinitis
<i>Nairovirus</i>	Nairobi sheep disease virus	Eastern Africa	Ticks	Sheep, goats	Hemorrhagic enteritis	Mild febrile illness
	Crimean-Congo hemorrhagic fever virus	Africa, Asia, Europe	Ticks	Sheep, cattle, goats, humans	Mild if any	Hemorrhagic fever, hepatitis

(Continued)

TABLE 3.1 (Continued)

Genus	Virus	Geographic Distribution	Arthropod Vector	Target Host Species or Amplifier Host	Disease in Animals	Disease in Humans
<i>Bunyavirus</i>	Akabane virus	Australia, Japan, Israel, Africa	Mosquitoes, Culicoides	Cattle, sheep	Arthrogryposis, hydranencephaly	None
	Cache Valley virus	United States	Mosquitoes	Cattle, sheep	Arthrogryposis, hydranencephaly rarely	Very rarely congenital infection
	La Crosse and other California encephalitis group viruses	North America	Mosquitoes	Small mammals, humans	None	Encephalitis
<i>Hantavirus</i>	Hantaan virus	China, Russia, Korea	None	<i>Apodemus agrarius</i> (striped field mouse)	None documented	Hemorrhagic fever with renal syndrome
	Puumala virus	Scandinavia, Europe, Russia	None	<i>Clethrionomys glareolus</i> (bank vole)	None documented	Hemorrhagic fever with renal syndrome
	Seoul virus	Worldwide	None	<i>Rattus norvegicus</i> (Norway rat)	None documented	Hemorrhagic fever with renal syndrome
	Sin Nombre virus and other New World hantaviruses	The Americas	None	<i>Peromyscus maniculatus</i> (deer mouse) and other reservoir rodent species	None documented	Hantavirus pulmonary syndrome

Genus *Phlebovirus*

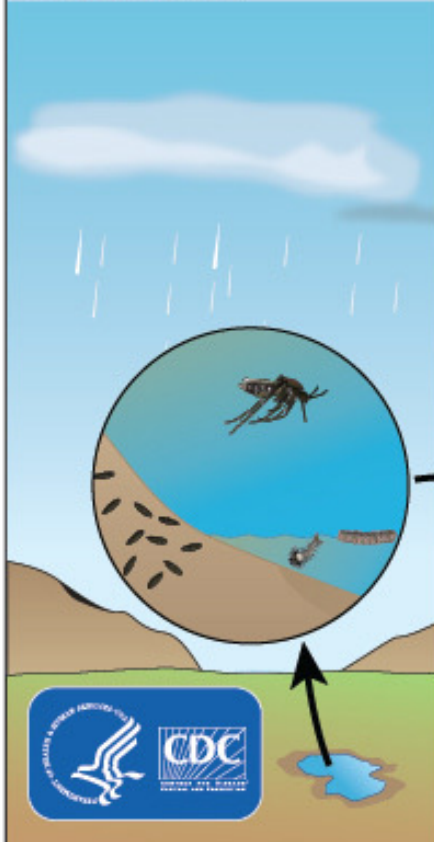


- Rift valley fever virus
 - Geographic distribution: Africa
 - Host: แกะ โค กระบือ และคน จัดเป็น zoonosis
 - Vector (พาหะ): ยุง (*Culex*, *Aedes*) หรือผ่านทาง oropharynx
 - Incubation period (ระยะฟักตัว): 30-72 ชั่วโมง
 - Symptoms (อาการ): ไข้ เบื่ออาหาร อาเจียน ท้องเสียปนเลือด น้ำมูกข้นเป็นหนอง แห้งและลูกตาย
 - Pathology (พยาธิสภาพ): เนื้อตายที่ตับ ม้ามโต จุดเลือดออกที่กระเพาะอาหาร ลำไส้ และเยื่อหุ้มสมองอักเสบ
 - Prevention and control (การป้องกันและควบคุม): ให้วัคซีนและควบคุมพาหะ
 - วัคซีนเชื้อเป็นห้ามบริหารให้ในสัตว์ตั้งท้องเพราะจะทำให้เกิดการแท้ง

Rift Valley Fever (RVF) virus ecology

Enzootic Cycle

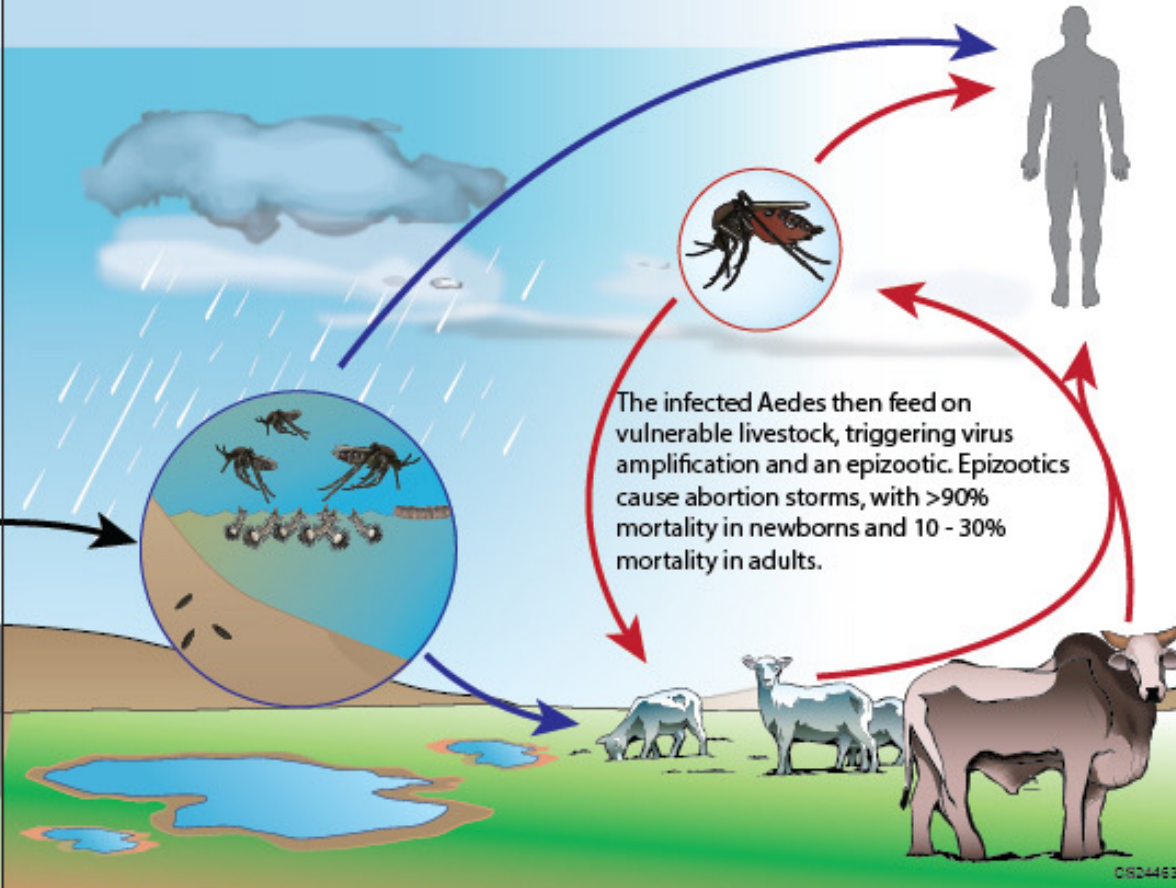
Local enzootic transmission of RVF occurs at low levels in nature during periods of average rainfall. The virus is maintained through transovarial transmission from the female Aedes mosquito to her eggs and through occasional amplification cycles in susceptible livestock.



Epizootic-Epidemic cycle

Abnormally high rainfall and flooding stimulate hatching of the infected Aedes mosquito eggs, resulting in a massive emergence of Aedes, including RVF virus-infected Aedes.

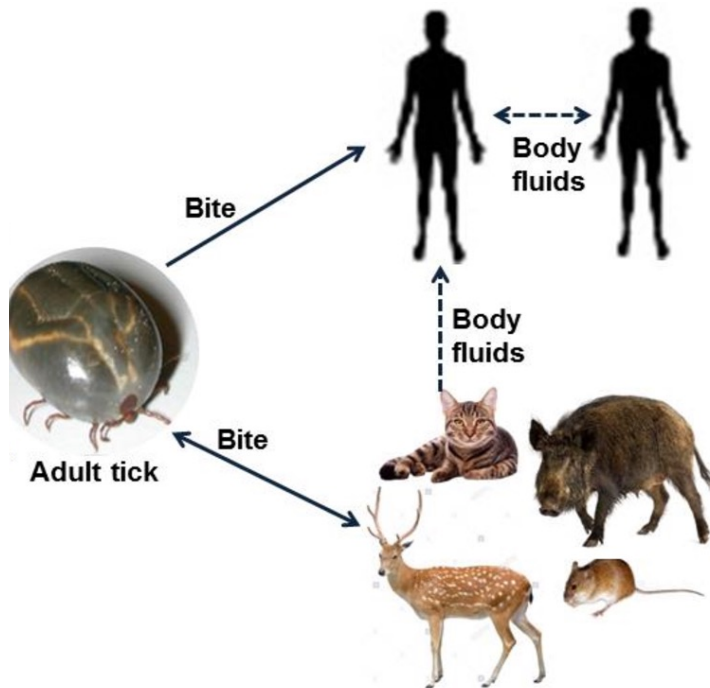
Secondary vectors include other mosquito genera such as Culex, which can pass on the virus to humans and animals, producing disease. Human exposure to viremic livestock (mostly small ruminants) blood and tissue can occur during slaughtering or birthing activities.



Diagnosis

- Rapid diagnosis: Reverse-transcriptase-polymerase chain reaction (RT-PCR) assays
- Confirmed by virus isolation by intracerebral inoculation of mice or in cell culture
- Serologic diagnosis:
 - Immunoglobulin (IgM) capture enzyme immunoassay on single acute sera
 - Enzyme immunoassays (ELISA), Neutralization test or Hemagglutination-inhibition assays on paired sera from surviving animals
 - เพื่อตรวจหาระดับ antibody titer ในสัตว์ที่รอดชีวิต

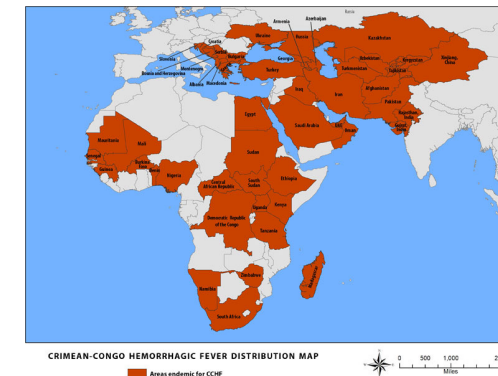
Severe fever with thrombocytopenia syndrome virus (a novel *Phlebovirus*)



- Severe Fever with Thrombocytopenia Syndrome (SFTS) is a rare case but emerging tick-borne viral disease that is present in Thailand
 - Although initially thought to be confined to East Asia (China, Japan, South Korea), retrospective studies and surveillance show that SFTSV has been circulating quietly in Thailand since at least 2012
- Animal reservoirs: studies show that local animal populations including dogs and cats carry antibodies for SFTSV, indicating the virus is spread locally among companion animals and ticks in various regions
- Symptoms (in human): sudden onset of fever, muscle pain, severe abdominal pain, vomiting, diarrhea, found leukopenia and thrombocytopenia
 - In severe cases, multi-organ failure or hemorrhagic symptoms with a high fatality rate (10-30%)

Genus *Nairovirus*

- Crimean-congo hemorrhagic fever virus (CCHF)
 - Geographic distribution: Africa, Asia and Europe
 - การติดต่อผ่านเห็บพาหะ (Ixodid ticks, especially those of the genus, *Hyalomma*)
 - สามารถก่อโรคได้ทั้งในสัตว์และในคน จัดเป็น zoonosis
 - อาการ: ไข้ อ่อนเพลีย มีจุดเลือดออกใต้ผิวหนัง ทางเดินอาหาร ระบบสืบพันธุ์และทางเดินปัสสาวะ ตับอักเสบ หัวใจและระบบประสาทถูกทำลาย
 - Diagnosis: enzyme-linked immunosorbent assay (ELISA), real time RT-PCR, virus isolation
 - Prevention: inactivated, mouse-brain derived vaccine → used on a small scale in Eastern Europe (no safe and effective for human use)





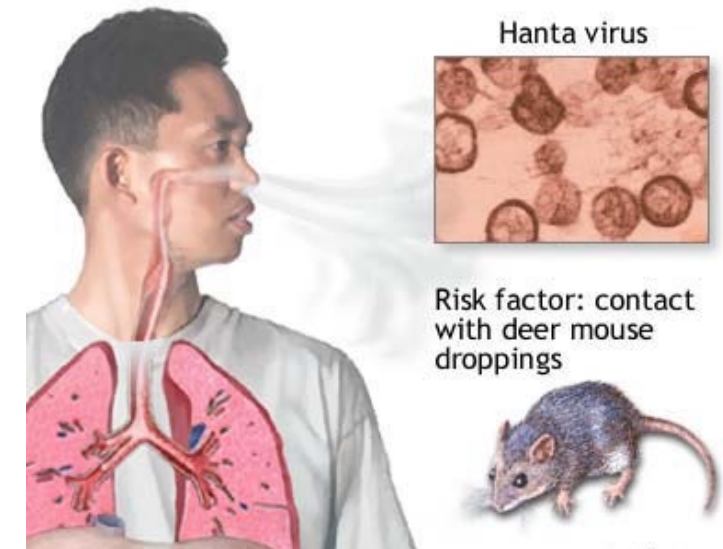
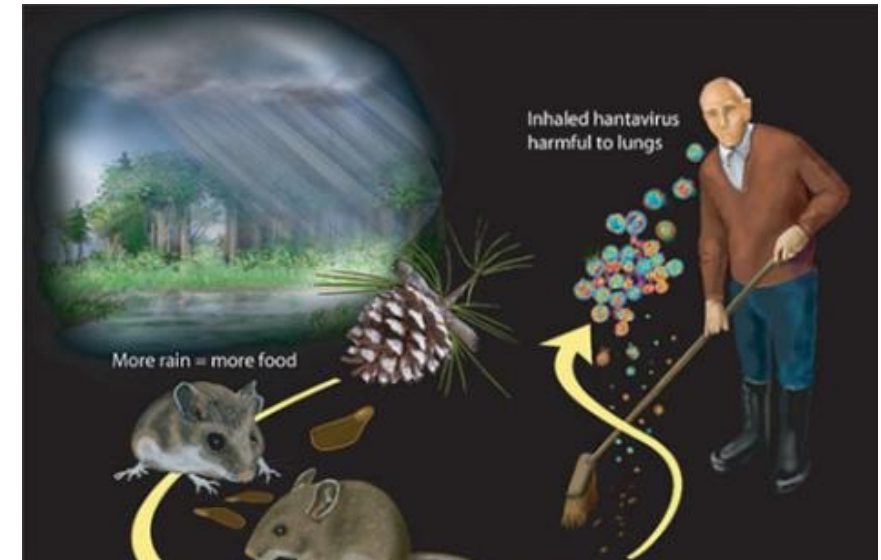
Genus *Orthobunyavirus*

- **Akabane virus**
 - Distribution: Africa, Asia and Australia
 - ก่อโรค Congenital arthrogryphosis-hydranencephaly syndrome
 - ติดต่อผ่านยุงพาหะและแมลงดูดเลือดพวก *Culicoides* ทำให้เกิดโรคในโค แพะ แกะ
 - อาการและพยาธิสภาพ: สมอและไขสันหลังอักเสบ และกล้ามเนื้ออักเสบใน fetus, ในรายที่ติดเชื้อรุนแรงตัวอ่อนจะตายและแท้งออกมา ตัวที่รอดชีวิตจะเกิด arthrogryphosis (ข้อยึดติด) และ hydranencephaly (เนื้อเยื่อสมองบางมีน้ำคั่งมาก)



Genus *Hantavirus*

- Hantaan virus
 - Geographic distribution: Asia and Europe
 - ก่อให้เกิดโรค Hemorrhagic fever with renal syndrome (HFRS) ในคน
 - ติดต่อผ่านหนู (rodent): *Apodemus agrarius*
 - อาการ: ไข้สูง ปวดหัว มีจุดเลือดออก ไตวายเฉียบพลันพร้อมอาการช็อค
 - การศึกษาในไทย:
 - พ.ศ.2528 Dr. Elwell และคณะ สามารถเพาะแยกไวรัสได้จากชิ้นเนื้อปอดของหนู 2 สายพันธุ์ ได้แก่ *Rattus norvegicus* และ *Bandicota indica*
 - พบว่าคนไทยที่ไม่มีอาการป่วยด้วยโรค HFRS มีแอนติบอดีต่อ Hantavirus อยู่ในช่วงระหว่าง 1-33%



Thailand Remains Hantavirus-Free as health Officials Tighten Surveillance at Entry Points

12/05/2026 | 1,194 |

Good news! Thailand has not recorded any hantavirus cases, and the overall public health risk remains low. But the Ministry of Public Health is stepping up monitoring at airports, ports, and border checkpoints, particularly for travelers arriving from high-risk regions, as a precautionary measure.

According to the spokesperson for the Department of Disease Control, Dr. Jurai Wongsawat, Hantavirus is carried by rodents such as field and house rats. It spreads mainly through inhaling dust contaminated with rodent urine or droppings in poorly ventilated spaces, not through person-to-person contact, which is extremely rare.

Early symptoms can resemble the flu. In severe cases, respiratory complications may develop. While some strains, particularly those found in parts of South America, have reported fatality rates ranging from 35% to 50%, those variants are not present in Thailand.

How to stay safe?

- Avoid contact with rodents
- Keep your home and workplace clean
- Store food in sealed containers



References

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